Project Report - ECE 176
ROS 2 and Gazebo-Based Color Sorting Pipeline

Ruochen Li
Department of Mechanical and Aerospace Engineering
PID: A17370336

Abstract

This project presents a complete robotic sorting pipeline in simulation, where a robot identifies the color category of a block and places it into the corresponding target area. The system is implemented in ROS 2 and Gazebo, while MoveIt 2 is used for motion planning and execution. The final task is to distinguish between two colored blocks and sort them into different locations.

The motivation of this project is to study how perception and manipulation can be integrated into a coherent robotic system. Even for a relatively simple task such as color-based sorting, the robot still needs to complete several connected stages: perceiving the target object, assigning a symbolic class label, mapping the label to a placement destination, and executing a collision-aware pick-and-place motion. The main purpose of this project is therefore not only to classify object colors, but to build a full perception-to-action loop that works reliably in simulation.

The final system demonstrates that two differently colored blocks can be successfully classified and placed separately. This result shows that the proposed pipeline provides a functional baseline for future extensions such as more object categories, stronger visual models, and more flexible pose-aware grasping.

1 Introduction

Robotic sorting is a representative task that combines perception, decision making, and manipulation. A robot must interpret sensory information, determine how an object should be categorized, and then physically move the object to the correct destination. Although each of these components can be studied independently, building a complete end-to-end system remains challenging because errors in one stage can affect all later stages.

The goal of this project is to build a compact but complete sorting pipeline in simulation. In the final version of the project, the robot observes colored blocks in the scene, classifies them into one of two color categories, and places them into different target locations. The system is implemented in ROS 2 and Gazebo, with MoveIt 2 handling motion planning and execution.

A major motivation for choosing this problem is that it is simple enough to be implemented robustly within one course project, while still requiring the integration of several important robotics subsystems. Even for a simplified task such as color-based block sorting, the full system must still solve perception, task mapping, motion planning, and execution consistently. This makes the project a good example of a practical robotics workflow, where system integration is just as important as the performance of any single component.

Another reason this problem is meaningful is that sorting is a common task in industrial and service robotics. In many real applications, robots need to detect different object categories and place them into separate containers or workspaces. While the final task in this project is intentionally simplified to two colors, the same system structure can later be extended to more classes, more complicated scenes, and more advanced perception modules.

The main contribution of this project is the successful implementation of an end-to-end color sorting pipeline. The final system can distinguish between two colored blocks and place them separately according to their predicted category. This demonstrates a complete perception-to-action loop in simulation and establishes a baseline for future work on more complex object classification and manipulation tasks.

2 Related Work

This project is related to visual classification, robot simulation, and motion planning. Lightweight classifiers such as MobileNetV3 are relevant because they map visual input to semantic labels. Gazebo is relevant because it provides a repeatable simulation environment for testing robotic systems. ROS 2 and MoveIt 2 are relevant because they support modular communication, motion planning, and execution.

Unlike prior work that focuses mainly on perception or planning alone, this project emphasizes system integration. Its main contribution is a complete color-based sorting pipeline that connects classification results to robotic pick-and-place actions in simulation.

3 Method

3.1 System Overview

The proposed method is a modular color sorting pipeline implemented in ROS 2. The system contains four main stages:

1. **Scene generation in Gazebo:** colored blocks, robot arm, and placement targets are placed in a simulation environment.
2. **Color classification:** the perception module identifies the color category of the target block.
3. **Task mapping:** the predicted color class is mapped to a predefined placement target.
4. **Motion planning and execution:** MoveIt 2 generates and executes a pick-and-place trajectory.

This modular design separates perception from robot control. As a result, the system is easier to debug, test, and extend. If a stronger visual classifier is added in the future, it can replace the current perception logic without requiring the rest of the sorting pipeline to be redesigned.

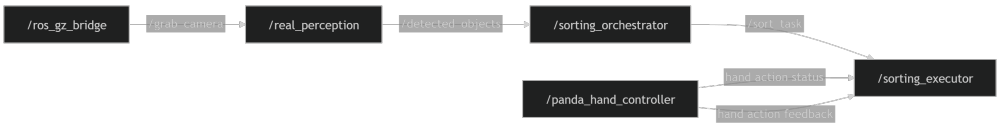

Figure 1: ROS 2 node graph of the final sorting pipeline. Camera images are bridged from Gazebo to the `/grab_camera` topic, processed by the `/real_perception` node, converted into detected object messages for `/sorting_orchestrator`, and finally sent as sorting tasks to `/sorting_executor`.

3.2 ROS 2 Node Structure

The final implementation follows a ROS 2 pipeline structure in which different functional modules communicate through well-defined interfaces. Conceptually, the system contains:

- a **perception-side module** that determines the color class of the target block,
- a **task mapper** that converts the predicted class into a target placement region,
- a **sorting executor** that performs the pick-and-place motion using MoveIt 2,
- the simulation and control infrastructure provided by **Gazebo**, **ROS 2 Control**, and **MoveIt 2**.

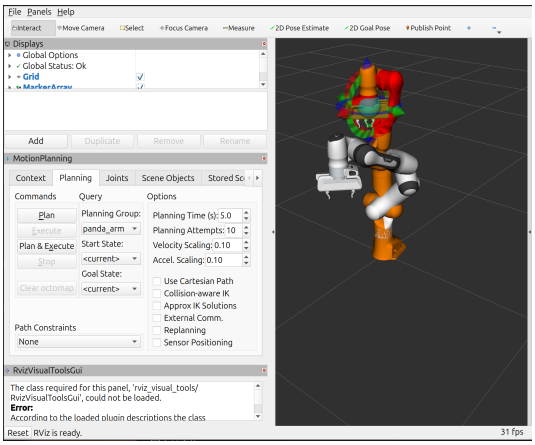

Figure 2: Enter Caption

The information flow is straightforward. Once a block is identified as belonging to one color category, the system assigns it to one placement area. If the block belongs to the other color category, it is assigned to a different placement area. The executor then performs a motion sequence to move the object from its current location to the assigned destination.

Figure 1 shows the ROS 2 communication structure of the final system. The pipeline is organized into modular nodes for perception, task mapping, and execution. Camera images are bridged from Gazebo into ROS 2, processed by the perception node, and then passed through the orchestration and execution stages to complete the sorting task.

3.3 Perception Module

The perception stage is responsible for distinguishing between two different colored blocks. Instead of solving a general object recognition problem with many classes, this project focuses on a simpler binary visual classification task. This simplification is appropriate for a course project because it allows the full robotic pipeline to be implemented and tested robustly.

The input to the perception stage is the visual observation or scene information associated with the target block. The output is a symbolic color label. This label serves as the interface between perception and action. In other words, the perception module transforms low-level visual information into a high-level decision that the rest of the robot system can use.

Even though this perception problem is simple, it is still important because it demonstrates how semantic information can be incorporated into a manipulation pipeline. The same design could later support more categories, richer visual features, or learned models.

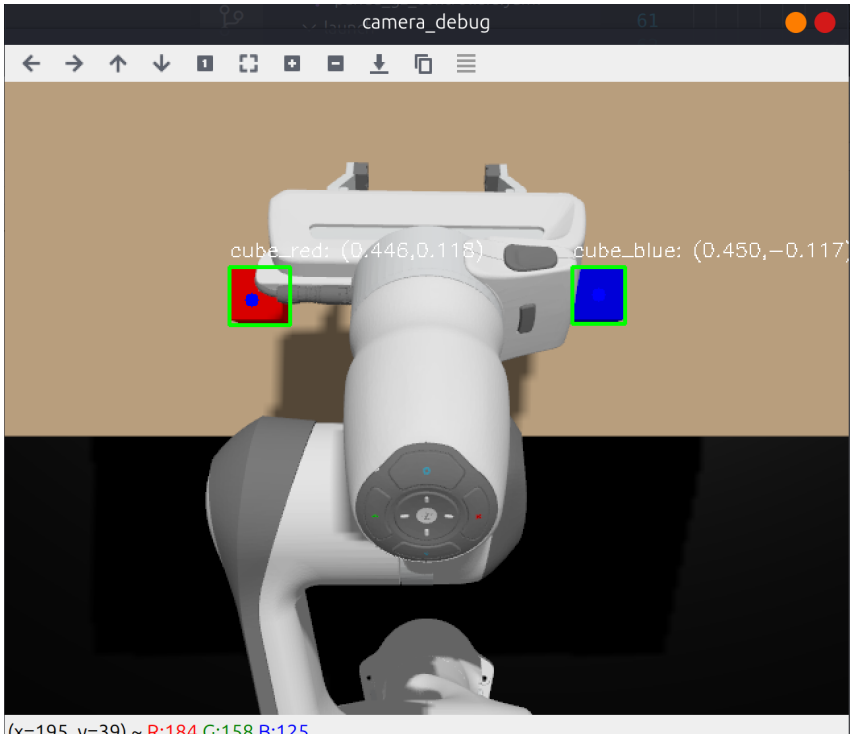

Figure 3: Camera-based perception result. The system correctly identifies the two colored blocks and assigns the corresponding labels, providing the semantic input for downstream sorting.

As shown in Figure 3, the perception module correctly distinguishes the two colored blocks and outputs the corresponding labels for downstream task mapping.

3.4 Task Mapping

Once the block color is predicted, the system assigns the block to a corresponding target area. For example, one color is mapped to one placement location, while the other color is mapped to another location. This stage converts a perception result into a symbolic robot task.

Task mapping is intentionally simple in this project, but it plays an important role in the overall architecture. It provides a clear separation between "what the object is" and "what the robot should do." This makes the system modular and interpretable. If the sorting rule changes in the future, the mapping logic can be updated without changing the perception module or the motion planner.

3.5 Motion Planning and Execution

After receiving the classification result and target location, the sorting executor triggers a pick-and-place sequence using MoveIt 2. The robot moves to a pregrasp pose, grasps the target block, lifts it, transports it to the assigned destination, and places it in the correct area.

A typical motion sequence in the system can be summarized as:

$$\text{pregrasp} \rightarrow \text{grasp} \rightarrow \text{lift} \rightarrow \text{transport} \rightarrow \text{place} \rightarrow \text{retreat}.$$

MoveIt 2 is used for trajectory generation and execution in simulation. This allows the project to focus on high-level integration between perception and manipulation rather than low-level controller design. Since the project is evaluated in a controlled simulated environment, the use of a structured pick-and-place pipeline provides a practical and stable solution.

In robotics, motion planning is often the stage where many integration failures become visible. Even when perception and symbolic reasoning are correct, the robot still needs a valid and reachable path

to complete the task. Therefore, the execution stage is essential for validating that the whole system works as an end-to-end robotic pipeline.

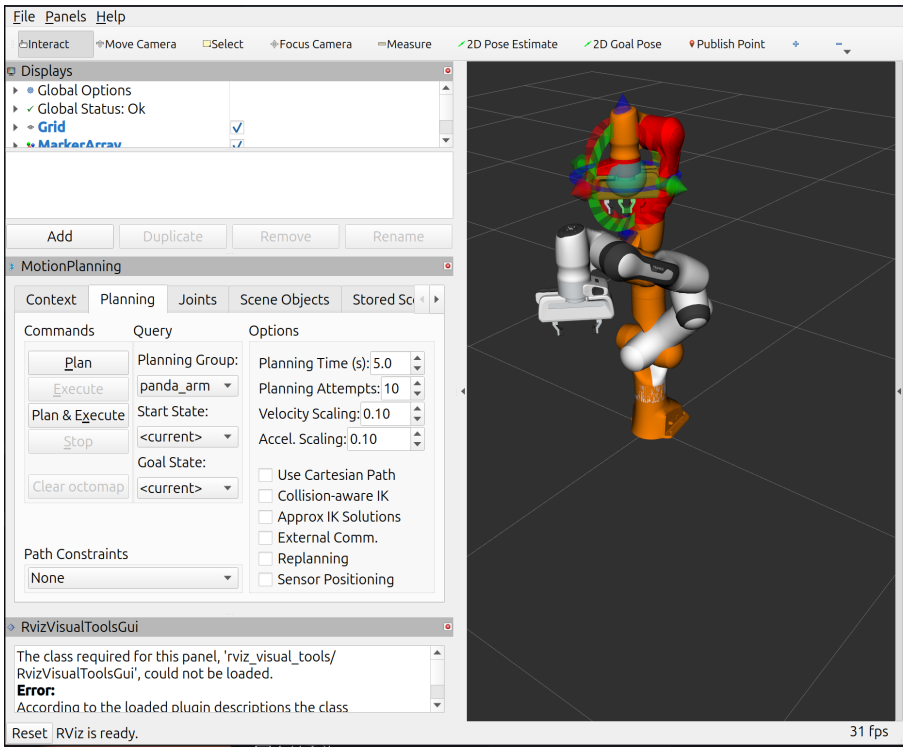

Figure 4: Motion planning interface in RViz. MoveIt 2 is used to generate and execute robot arm trajectories for the pick-and-place stage of the sorting pipeline.

Figure 4 shows the MoveIt 2 planning interface used for trajectory generation and execution in the manipulation stage.

3.6 Why This Method Was Chosen

The main reason for choosing this method is that it provides a manageable but complete robotics problem. A color-based two-class sorting task is simple enough to be implemented reliably, but still requires correct coordination between classification, symbolic decision making, and motion execution. This makes it an effective baseline for studying robotic system integration.

Another advantage of this method is clarity. Because the perception problem is relatively simple, it is easier to identify whether failures come from classification, task mapping, or motion planning. This is especially helpful in a course project setting, where interpretability and stable progress are important.

Overall, the method was chosen because it balances feasibility and completeness. It is complex enough to demonstrate a real robotic workflow, but simple enough to be finished and validated within the scope of the project.

4 Experiments

4.1 Simulation Setup

All experiments were conducted in a ROS 2 + Gazebo simulation environment. The scene contains a robot manipulator, colored blocks, and predefined target locations for sorting. The final task is to classify two different colored blocks and place them into separate locations.

The system is evaluated as a complete robotic pipeline rather than as an isolated vision model. For this reason, the most important experimental question is not only whether the block color is recognized correctly, but whether the robot can complete the final sorting task successfully.

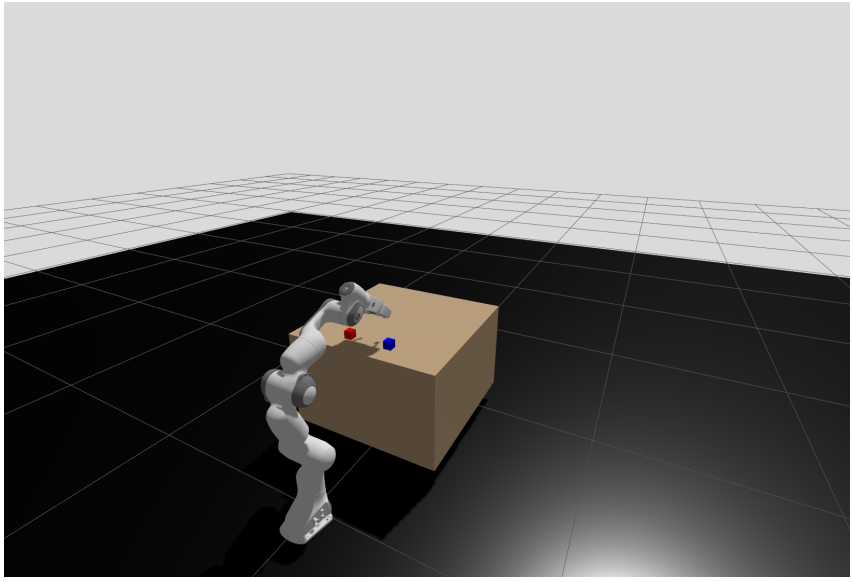

Figure 5: Gazebo simulation environment used in the experiments. The scene contains a robot manipulator and two colored blocks placed on the workspace for sorting.

Figure 5 illustrates the Gazebo scene used to evaluate the proposed color sorting pipeline.

4.2 Dataset and Input Format

The perception input consists of visual observations of colored blocks in simulation. Each sample corresponds to a target block associated with one of two color labels. The classification result is passed to downstream modules in the form of a symbolic class label.

In a broader machine learning sense, the input format can be viewed as a simple two-class dataset, where each visual observation belongs to one of two color categories. However, the purpose of this project is not to report an extensive standalone vision benchmark. Instead, the main focus is to verify that the classification result can be used correctly by the robotic sorting system.

4.3 Evaluation Metrics

The project is evaluated at the system level using three metrics. First, **classification accuracy** measures whether the color of the target block is identified correctly. Second, **overall sorting success rate** measures whether the robot completes the full pipeline successfully, including classification, motion planning, grasping, and final placement. Third, **failure breakdown** records the cause of unsuccessful trials, which helps determine whether failures come from perception or manipulation.

These metrics are appropriate because the goal of the project is not only to recognize object color, but to complete a full perception-to-action sorting task. A correct classification is meaningful only if it can be converted into a successful robot action.

4.4 Results

The final system was tested in 10 trials. In all 10 trials, the robot correctly identified the color of the target block, resulting in a classification accuracy of 100%. Among the 10 trials, 7 trials successfully completed the full sorting task, while 3 trials failed because inverse kinematics could not generate a feasible motion. Therefore, the overall sorting success rate was 70%, and the IK failure rate was 30%.

These results show that the perception module worked reliably in all tested cases. The system consistently assigned the correct semantic label to each block and mapped that label to the correct placement target. The unsuccessful trials were caused by motion feasibility issues rather than by classification errors, indicating that the main bottleneck of the current system lies in manipulation robustness.

Overall, the experiments validate the proposed ROS 2 + Gazebo + MoveIt 2 pipeline. The robot was able to distinguish between two differently colored blocks and sort them into separate target locations, demonstrating a complete perception-to-action workflow in simulation.

Table 1: Quantitative results of the final color sorting system.

Metric	Result
Total trials	10
Classification accuracy	100% (10/10)
Successful sorting trials	7
Overall sorting success rate	70% (7/10)
Failed trials due to IK failure	3
IK failure rate	30% (3/10)

4.5 Discussion

The experimental results suggest that perception was not the limiting factor in the final system. Since all 10 trials were classified correctly, the color recognition component was sufficiently reliable for this task. The primary source of failure came from inverse kinematics, which prevented the robot from completing the manipulation sequence in several trials.

This observation is important because it shows that, even in a relatively simple sorting problem, the main challenge of an end-to-end robotics pipeline may come from motion execution rather than visual recognition. In other words, the semantic part of the task was solved successfully, but the physical execution stage still requires further improvement.

There are several directions for future work. First, the manipulation stage can be made more robust by improving grasp pose generation, reachability checking, or trajectory planning. Second, the perception module can be extended from two-color classification to more object categories or more complex visual conditions. Third, the pipeline can be tested in more challenging environments, such as cluttered scenes or settings with higher pose variability. These improvements would make the system more general while preserving the modular structure established in this project.

5 Supplementary Material

A supplementary presentation video accompanies this project. The video introduces the motivation, system design, and final experimental results of the project. In particular, it demonstrates that the robot can correctly distinguish between two colored blocks and attempt to place them into different target locations according to the classification result. The video serves as a visual summary of the complete perception-to-action sorting pipeline implemented in simulation.

References

[1] Andrew Howard, Mark Sandler, Grace Chu, Liang-Chieh Chen, Bo Chen, Mingxing Tan, Weijun Wang, Yukun Zhu, Ruoming Pang, Vijay Vasudevan, et al. “Searching for MobileNetV3.” arXiv preprint arXiv:1905.02244, 2019.

[2] Josh Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and Pieter Abbeel. “Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World.” arXiv preprint arXiv:1703.06907, 2017.

[3] MoveIt 2 Documentation. ROS 2 motion planning framework documentation.

[4] Gazebo Documentation. Robot simulation and ROS integration documentation.

[5] ROS 2 Documentation. Communication framework for modular robotic systems.